

Case study

Customer Funnel Analysis | Python, Power BI

1. Business Problem:

The business was facing low conversion rates due to significant user drop-offs in the funnel, especially between browsing and purchase stages. There was no clear visibility into where the highest drop-offs were occurring across regions, channels, and time periods, leading to missed revenue opportunities.

2. Objective:

Identify the key stages in the funnel where users drop off from browsing to purchase and quantify the impact on overall conversion rate.

Analyze how region, channel, and time influence conversion efficiency to find the main drivers of revenue leakage.

3. Tools & Project Links:

Python : [Link](#) | Power BI: [Link](#)

4. Analysis:

- Performed EDA
- Built funnel analysis
- Performed channel performance analysis
- Conducted regional analysis
- Performed device and product category analysis
- Conducted time-based analysis
- Performed user funnel performance analysis
- Performed funnel drop-off diagnostics

5. Key Insights

- Biggest drop-off observed at **Purchase stage (69.35%)**, indicating major leakage before final conversion
- **Organic channel performed best** with highest conversion rate (11.19%), showing strong intent-driven traffic
- **South region showed highest conversion rate (11.25%)**, indicating better regional engagement efficiency
- **Beauty category recorded highest conversion (11.43%)**, suggesting strong product-market fit in this segment

6. Recommendations

- Improve user experience at the Purchase stage to reduce high funnel drop-offs
- Increase investment in the Organic channel due to its highest conversion performance
- Replicate successful South region strategies across underperforming regions
- Promote Beauty category products to improve overall conversion performance
- Implement cart abandonment recovery strategies to recover 50.08% of users
- Identified a potential revenue recovery opportunity of **\$3.84M** through cart abandonment reduction strategies